

Tesla Inc. (TSLA)

Financial Analysis & Equity Valuation

Investment Analysis | Neil Changela

Company Overview – Tesla Inc.

Background & Business Model

- Founded 2003 by Eberhard & Tarpenning; Elon Musk joined 2004
- HQ: Austin, TX | Listed on NASDAQ since 2010
- ~125,000 employees
- Direct-to-consumer sales model – no dealerships
- Makes EVs, energy storage (Powerwall, Megapack), and sells FSD software subscriptions

Key Revenue Segments (2025)

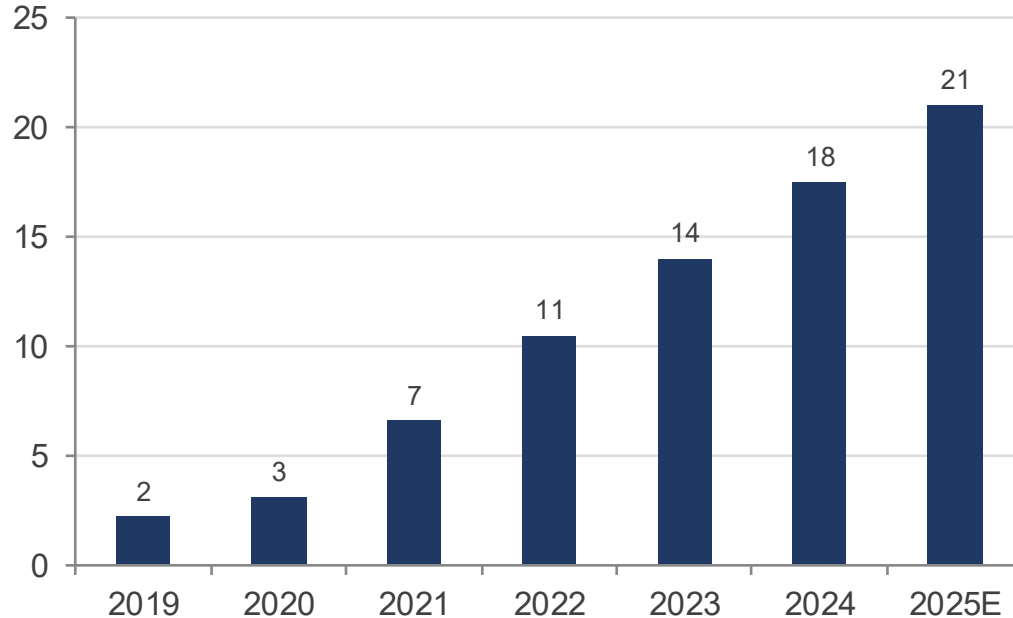
Segment	Revenue	% of Total
Automotive	\$69.5B	73%
Energy Storage	\$12.8B	14%
Services & Other	\$12.5B	13%

Strategic Positioning

- Vertically integrated: batteries → software → cars
- 4 Gigafactories: Texas, California, Germany, China
- ~75,000+ Superchargers globally
- Premium brand expanding into mass market
- Model Y was the world's best-selling car in 2025
- 2025: launched Robotaxi service in Austin (June 2025)
- Model S & X discontinued Jan 2025 – lines now building Optimus robots

Industry Analysis – Global EV Market

Global EV Sales (millions of units)



Key Drivers

- Market projected ~\$700B by 2030 (CAGR ~26%)
- Govt incentives: IRA credits (US), EU Green Deal
- Battery costs now ~\$115/kWh vs \$1,200 in 2010
- Growing consumer preference for EVs globally
- China = 63% of all global BEV sales (2025)

Key Risks

- U.S. EV tax credit (\$7,500) eliminated in Q3 2025
- Trump-era tariffs raising manufacturing costs
- Intensifying competition from BYD and legacy OEMs
- High interest rates reduce EV affordability
- Brand/reputational risks

Competitive Landscape

Company	2025 Deliveries	Revenue	Gross Margin	Key Note
Tesla (TSLA)	1.64M (↓9%)	\$94.8B	~18%	Pure EV leader; Robotaxi launched Austin
BYD (China)	~4M+ BEV+PHEV	~\$100B+	~22%	#1 global EV seller by volume; growing in EU
Ford (F)	~150K EVs	\$185B	~10%	Model e EV segment still losing money
GM	~200K EVs	\$187B	~13%	Ultium platform ramping; fastest growing in US
Rivian	~51K	~\$5.1B	~10%	Amazon partnership; still pre-profit

Key Takeaways

- Tesla still leads U.S. EV market but lost European market share (17.9% → 9.3% in Q1 2025 vs Q1 2024)
- BYD surpassed Tesla in global EV volume — the biggest competitive threat Tesla faces
- GM is the fastest growing U.S. EV brand and is catching up quickly
- Tesla's real advantage is software (FSD) and Supercharger network — no competitor has matched either

Strengths & Weaknesses

✓ Strengths

- First-mover advantage in premium EV market
- Highest gross margins among pure EV makers (~18%)
- Proprietary FSD = potential high-margin recurring revenue
- Supercharger network (50,000+ chargers) – massive moat
- Strong balance sheet: \$44.1B cash at end of 2025
- Robotaxi service now live in Austin (June 2025)
- Model Y – world's best-selling car in 2024

✗ Weaknesses

- Heavy Elon Musk dependence (key-person risk)
- Consumer backlash from Musk's political activities hurt 2025 sales
- 2 consecutive years of delivery declines (2024 and 2025)
- Operating margin dropped from 16.8% (2022) to 4.6% (2025)
- FSD not yet fully autonomous – regulatory approval still pending
- Discontinued Model S/X – smaller lineup vs. traditional OEMs
- Manufacturing quality issues and recalls

Stakeholder Analysis

Suppliers

- CATL & Panasonic supply batteries – lithium/cobalt prices are key risk
- Tariffs on Chinese battery materials raising input costs in 2025
- Tesla moving to in-house 4680 cells to reduce dependency

Investors

- Top holders: Musk (~13%), Vanguard, BlackRock
- Expectations are very high – stock trades at 318x earnings (Apr 2026)
- Very volatile stock – Beta ~1.91; not for risk-averse investors

Customers

- Early adopters → mass market; price cuts in 2023-24 upset early buyers
- Brand trust hit by Musk's politics – protests at showrooms in 2025
- FSD subscribers create sticky recurring revenue relationship

Government

- U.S.: EV tax credit (\$7,500) eliminated Q3 2025 – hurt demand
- EU: Carbon regulations still favor EVs long-term
- NHTSA actively investigating Autopilot and FSD safety

Future Outlook & Economic Sensitivity

Growth Catalysts

- Robotaxi (Cybercab) – launched in Austin Jun 2025; safety monitors removed Jan 2026
- Affordable Model Q / entry-level EV targeted at ~\$25K – could unlock huge market
- Optimus robots – production lines installed in Fremont 2025; potential blue-sky revenue
- Energy storage (Megapack) – revenue up 67% in 2024, still growing fast
- FSD unsupervised rolling out – if approved widely, creates a recurring high-margin revenue stream
- Expansion in Southeast Asia, India still largely untapped

Economic Cycle Sensitivity

Expansion

Strong deliveries, good margins

Recession

EVs are discretionary – buyers delay \$40K+ purchases

High Rates

Financing gets expensive; kills EV affordability

Low Rates

Cheap financing = strong EV demand tailwind

Trade War

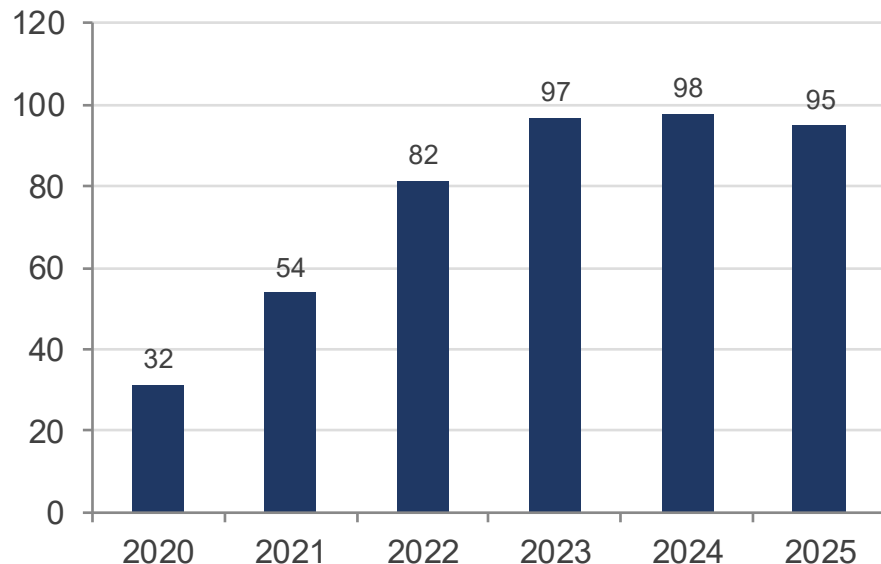
Tariffs raise battery/input costs (happening now in 2025)

Oil Spike

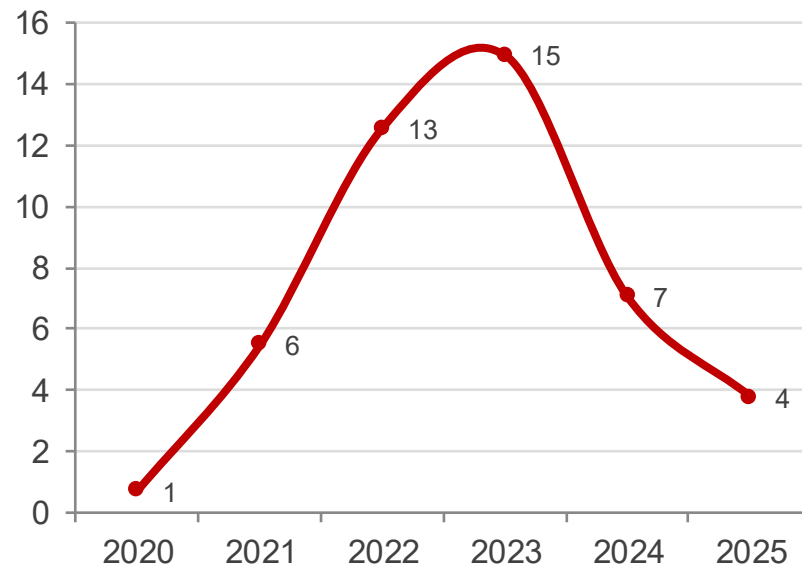
Makes EVs relatively cheaper vs gas cars

Financial Analysis – Revenue & Net Income

Total Revenue (\$ Billions)



Net Income (\$ Billions)



\$94.8B

2025 Revenue
(↓3% YoY – first ever decline)

3×

Revenue growth
(2020 → 2025)

\$3.8B

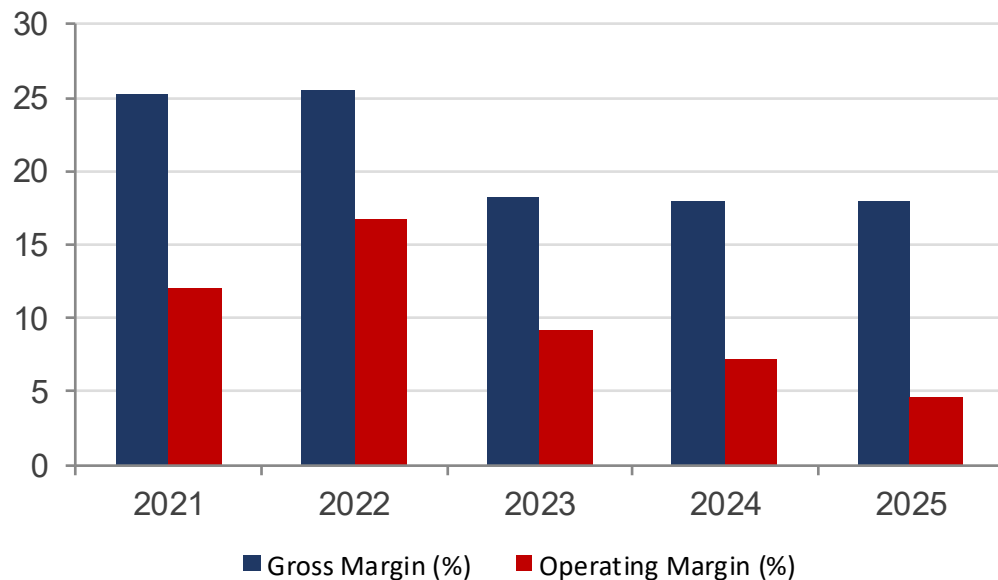
2025 Net Income
(↓46% from 2024)

\$44.1B

Cash & Investments
(record high, end 2025)

Profitability Analysis

Gross & Operating Margins (%)



Key Ratios (2025 vs 2022 Peak)

	2022	2025
Gross Margin	25.6%	18.0%
Op. Margin	16.8%	4.6%
Net Margin	15.4%	4.0%
ROE	~28%	~4.9%
EPS (GAAP)	\$3.62	\$1.08
FCF	\$7.6B	\$6.2B

⚠ Key concern: Operating margin collapsed from 16.8% (2022) → 4.6% (2025). This is due to aggressive price cuts to stay competitive with BYD, combined with rising operating expenses. FCF is still solid at \$6.2B – the cash generation is holding up better than accounting profits.

Key Financial Insights

Revenue Hit a Wall

Revenue grew nearly 4x from 2020 to 2024 (\$31B → \$97B) but fell in 2025 for the first time ever to \$94.8B (−3%). Automotive revenue is down 16% from its peak as Tesla faces competition and removed price cuts.

Margin Compression is Real

Operating margin: 16.8% (2022) → 4.6% (2025). Tesla cut vehicle prices ~20% in 2023 to keep volumes up vs. BYD. Margins have not recovered. This is the single biggest risk factor for investors.

Free Cash Flow Recovering

FCF bounced back to \$6.2B in 2025 from just \$3.5B in 2024. CapEx moderated from \$11.3B (2024) to ~\$8.5B as major Gigafactory construction wound down. Cash pile hit a record \$44.1B.

Non-Auto Segments Growing

Energy revenue: \$6B (2023) → \$10.1B (2024) → \$12.8B (2025). Services also hit \$12.5B. Non-auto revenue is now 27% of total and helps offset automotive weakness. Energy storage is becoming a real business.

DCF Valuation – Discounted Cash Flow

What is a DCF?

- DCF = estimate company value based on its future cash flows
 - Step 1: Project future Free Cash Flows (FCF)
 - Step 2: Pick a discount rate (WACC)
 - Step 3: Calculate Terminal Value
 - Step 4: Discount all cash flows back to today (PV)
 - Step 5: Divide by shares = implied price/share
- Compare result to market price → over or undervalued?

Our Assumptions (2025 base)

Base FCF (2025 actual)	\$6.2B
FCF Growth Years 1–5	15%/yr
FCF Growth Years 6–10	10%/yr
WACC	10%
Terminal Growth Rate	3%
Shares Outstanding	~3.2B

DCF Output

PV of FCF (Years 1–10)	\$72.3B
Terminal Value (PV)	\$213.4B
Enterprise Value	\$285.7B
Less: Net Debt	(\$7.2B)
Equity Value	\$278.5B
÷ Shares (3.2B)	
Implied Price / Share	~\$87
Current Price (Apr 2026)	~\$350

**⚠ DCF says ~\$87/share. Stock trades at ~\$350.
That's roughly 4× intrinsic value under base case assumptions.**

Comparable Valuation – P/E Multiples

Company	P/E (TTM)	EPS	Rev Growth	Gross Margin	Sector
Tesla (TSLA)	~318×	\$1.09	-3% (2025)	~18%	EV / AI
BYD	~18×	N/A	+22%	~22%	EV
Ford (F)	~8×	~\$1.80	+5%	~10%	Auto
GM	~5×	~\$10	+4%	~13%	Auto
Apple	~28×	~\$6.40	+4%	~46%	Tech
S&P 500	~22×	N/A	N/A	N/A	Market

What does this mean?

- If Tesla is valued like Ford/GM (8× P/E) → implied price = ~\$9/share. Obviously not fair – Tesla has better growth and tech.
- If Tesla is valued like a tech company (30× P/E) → implied price = ~\$33/share. Still way below \$350.
- At 318× P/E, investors are not paying for 2025 earnings – they're paying for the hope of FSD, Robotaxi, and Optimus robots.
- Bear case: EPS crashed from \$4.30 (2023) to \$1.09 (TTM) – the P/E expanded as earnings fell. That's risky.

Beta Analysis – Risk vs. The Market

What is Beta (β)?

- Beta measures how much a stock moves relative to the overall market
- $\beta = 1.0 \rightarrow$ moves exactly with the S&P 500
- $\beta > 1.0 \rightarrow$ more volatile than the market (amplified swings)
- $\beta < 1.0 \rightarrow$ less volatile (e.g. utility stocks)
- Formula: $\beta = \text{Cov}(\text{stock return}, \text{market return}) / \text{Var}(\text{market return})$

Beta Scale (examples)

Beta Range	Risk Level	Example
< 0.5	Very low	Duke Energy (utilities)
0.5–1.0	Low	Johnson & Johnson
1.0–1.5	Moderate	Apple (~1.2)
> 1.5	High	Growth/tech stocks
~1.91	Very High	Tesla

Tesla's Beta

$$\beta \approx 1.91$$

(5-year monthly, Yahoo Finance, April 2026)

What $\beta \approx 1.91$ means:

- Market up 10% $\rightarrow$ Tesla typically up ~19%
- Market down 10% $\rightarrow$ Tesla typically down ~19%
- In Q1 2025, Tesla fell 36% in a single quarter
- Not suitable for risk-averse or income investors
- Fine for high-risk-tolerance, long-horizon investors

CAPM – Required Rate of Return

$$E(r) = R_f + \beta \times (R_m - R_f)$$

R_f (Risk-Free Rate)

4.5%

10-yr US Treasury yield

β (Beta)

1.91

Tesla's systematic risk

R_m (Market Return)

10%

Historical S&P 500 avg

R_m – R_f (Premium)

5.5%

Equity risk premium

$$E(r) = 4.5\% + 1.91 \times (10\% - 4.5\%) = 4.5\% + 10.5\% = 15.0\%$$

What this means:

- Investors should REQUIRE a 15% annual return from Tesla to be compensated for the risk they're taking
- If Tesla can't deliver 15%+ annualized returns from today's price (~\$350), it's not worth the risk
- This is the hurdle rate – Tesla must grow earnings fast enough to clear this bar

Final Recommendation



HOLD



Bear Case (reasons to sell)

- DCF says ~\$87/share – stock at \$350 = 4× overvalued on fundamentals
- P/E of 318× is unjustifiable if EV margins keep falling
- 2 consecutive years of declining deliveries (2024 & 2025)
- Operating margin crashed from 16.8% → 4.6% in 3 years
- Musk brand damage hurt EU market share badly in 2025
- Beta of 1.91 = massive downside risk if market sells off



Bull Case (reasons to buy)

- Robotaxi live in Austin – first mover in autonomous driving
- FSD optionality not in DCF – could be worth trillions if it works
- Record cash pile: \$44.1B – funds all future bets
- Energy storage growing 67% YoY – potential standalone business
- Optimus robots = billion-dollar opportunity if it materializes
- FCF recovered to \$6.2B in 2025 – still generates cash

HOLD – fundamentals say overvalued but robotaxi/FSD optionality is real. Only suitable for high risk tolerance, long-term investors. Watch: Q1 2026 earnings (April 22, 2026).

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